

# 1 Section 1

## Real Time System Design for Autonomous Systems

Applications

- Smart IoT Monitoring

### **PAL: Privacy & Anonymity Lab**

Differential Privacy

Domain Registration Privacy under the GDPR

Investigate Privacy Concerns in emerging machine learning models

Investigating security issues in AI models via system logs

### **Bryan Donyanavard System Autonomy**

challenges

- environmental diversity
- platform complexity
- network variability

active research

- realtime environmental understanding
- operating systems and architecture

### **Hoa Vu Algorithms Research**

Streaming Algorithms

- problem: data arrives one at a time; too much to store
- answer questions with far less memory than the data itself
- given a stream of ip addresses, for example, how many distinct addresses have we seen.

classical algorithms

- assume no help (worst case analysis) but real world data has structure
- goal is to speed up algorithms
- improve performance

design and analyze algorithm design with practical machine learning

bridging rigorous algorithm design with practical machine learning

### **Yifan Zhang Security and Privacy in everyday software**

- hidden risks across software ecosystems
- software supplychains – mobile systems – connected devices

AI for finding subtle software bugs - 123 new bugs discovered across 154 python and c repositories on github

Deceptive advertising inside mobile apps

## **Sekeh Lab Foundations of Reliable Efficient and Robust AI driven systems in open world from theory to applications**

robot navigation in a dynamic world environment

physical AI and open world challenges

failure detection for robots

known unknowns and unknown unknowns: ai's blindspot problem

safety is an architecture

AI for healthcare

- post surgical wound care with multi-agent systems

### **Hajar Homayouni Data Science Lab**

Improving Data Quality and reliability

Improving interpretability of machine learning models to gain insights

- synthetic medical data generation

privacy preserving synthetic medical data generation

issue with ai: mimic real data - can reveal sensitive information

anomaly detection and explanation

- EDA of Medical Publications

### **Xin Zhang Urban AI Lab**

how can AI understand urban dynamics and human decisions, and turn that knowledge into trustworthy decisions

predicting urban dynamics

- how can we model complex spatial temporal dependencies to predict what will happen and how urban systems may evolve.

understanding human decisions

- how can ai learn how people make decisions from observed behavior.

structuring decision knowledge

- how can ai discover reusable skills and concepts that make complex decisions compositional and reasoning ready

Trustworthy Decision Making

- how can ai reason over learned decision knowledge to act effectively while respecting fairness, privacy, and human values

### **Performant Intelligent Computing with Evolving Systems Lab**

system development

- architecting and deploying diverse cloud systems

ADT Platform for Traffic Simulation

github: PiscestLab

Empirical Investigation

- employing timeline analysis to research performance anomalies

Intelligent Solutions

- designing strategies to mitigate performance variations

## **Machine Vision and Perception Lab Xiaobai Liu**

embodied AI

- cooking robots
- generalist robots

agent as policy research

whistle extraction

- deep whistle – patch based residual architecture - SAM Whistle – segmentation driven extraction

Homelessness Survey

- detection and geospatial mapping

## **Wei Wang**

uberize your wireless access

treat device as a human or as an agent

local wireless ecosystem: uberize token, pinch wireless access

intelligent wireless networks research from bits to semantics

intelligent wireless multimedia networks model as data

## **Shangpin Ren Integrating Cyber innov with the phys world**

Infrared Pen and App for Raynaud's Patients

AI assisted capsule endoscopy clinical review platform

virtual cell modeling from drug effects to assay responses

conditioned 3D Geometric Modeling for Dental Design

## **2 Section 2**